Boundary Completion for Quadratic Polylogarithm Feedback

Convergent inverse transseries, logarithmic resonance, and coefficient transitions

Research report prepared for the ProveIt project

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Abstract

We resolve the boundary-feedback question explicitly left open in the incoming report *Resonant Polylogarithm Transseries*. For the germ $v(0) = 0$ determined by $uv - v^2/2 = \mathrm{Li}_s(re^v)$, we prove the exact Taylor radius, unique dominant singularity, and leading coefficient law for every $s > 0$ and $u > 0$. For $s > 2$, the critical parameter is $U_c(s)^2 = 2\zeta(s) + \zeta(s-1)^2$. Above it we prove the previously conjectured $R^{-n}n^{-s}$ law, including all integer orders. At equality the three coefficient scales are $R^{-n}n^{-1-1/(s-1)}$ for $2 < s < 3$, $R^{-n}n^{-3/2}(\log n)^{-1/2}$ for $s = 3$, and $R^{-n}n^{-3/2}$ for $s > 3$, with explicit constants. An exact integral normal form yields convergent inverse charts in finitely many power and logarithmic coordinates. At order three an exact Lambert core gives every inverse-logarithmic coefficient correction; the first correction has a particularly simple constant. We also construct a convergent uniform chart as the order crosses three and prove the coefficient crossover as the feedback parameter crosses its critical value. The proofs use classical polylogarithm continuation, holomorphic inversion, singularity analysis, and a directly proved lattice local limit argument. The contribution is the completed nonlinear model, its exact constants and charts, and the resolution of a repository conjecture; the general stable-law and singularity-analysis mechanisms are credited to their antecedents. Reproducible algebraic and numerical checks accompany the proofs.

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1 Research question, source audit, and precise contribution

The nonlinear feedback equation

$$uv(r) - \frac{v(r)^2}{2} = \mathrm{Li}_s\big(re^{v(r)}\big), \qquad v(0) = 0, \qquad s > 0, \quad u > 0, \tag{1.1}$$

provides a concrete meeting point for polylogarithms and transseries. The polylogarithm supplies a boundary singularity whose exponent changes with s. Nonlinear inversion may reach that singularity before, after, or exactly when the ordinary inverse derivative vanishes. The equality case produces fractional powers, logarithms, and iterated logarithms; it is especially well suited to a precise distinction between a formal expansion and a convergent analytic chart.

The source report [1], Theorem `thm:feedback`, proves the case in which an interior characteristic point exists. Its first question in Section `sec:future` asks for the remaining boundary cases and prints the supercritical coefficient constant as an unproved candidate. We prove that candidate and all critical cases. This is a resolution of an explicitly stated project question, not a claim to settle a previously established open problem in the wider literature.

1.1 Snapshot and comparison sources

The repository was inspected at commit

`6207c3d40047fd8d1400a4d509fee71c9a3a8132`.

The review covered the two topic indexes, the incoming-report index, the five incoming ZIP articles, and the relevant critical-feedback and endpoint articles in the Transseries collection. It was a focused mathematical audit, not a line-by-line verification of the complete 552-page Polylogarithms manuscript or of every Transseries report. The source inventory and hashes are recorded in `provenance.json`.

Source	Result already available	Role in this report
Incoming feedback report [1]	Positive finite coefficients, exact interior radius and fold, and the boundary conjecture	Interior theorem credited; boundary conjecture proved and extended
Critical Hahn report [2]	Linear feedback with power tails, critical fractional inverses, stable coefficient laws, coupling crossover	General mechanism and comparison model
Gaussian endpoint report [3]	Uniform order crossover for eventually exact power tails, Lambert coordinates and local limits	Antecedent for the order–logarithm scaling
Other four incoming articles [4, 5, 6, 7]	Resonant order jets, ordinary polylogarithm inverse charts, complex critical values and folds	Prevents presenting those existing charts as new results

The nonlinear function arising here is

$$f_u(t) = u - \sqrt{u^2 - 2\,\mathrm{Li}_s(t)}. \tag{1.2}$$

Even when its coefficients have a power-law tail, they are not in general eventually an exact multiple of n^{-s}. Thus the existing theorems for a polylogarithm plus a finite polynomial do not by themselves verify this model's endpoint continuation or its correction constants. We give those proofs explicitly.

1.2 Classical inputs and novelty boundary

We use the classical convergent Jonquière expansion

$$\mathrm{Li}_s(\mathrm{e}^{-x}) = \Gamma(1-s)x^{s-1} + \sum_{k\geq 0} \zeta(s-k)\frac{(-x)^k}{k!}, \qquad 0 < |x| < 2\pi, \tag{1.3}$$

for nonintegral positive s, with a fixed branch of $\operatorname{Log} x$. At positive integers the colliding Gamma and zeta terms are combined before the parameter limit is taken. These are classical identities, recorded in DLMF §25.12 [8]. All displayed local series below are restricted to sufficiently small sectors within this convergence domain.

The conversion of local singular expansions into Taylor coefficient asymptotics is the singularity-analysis theory of Flajolet and Odlyzko [10]. The relation between implicit generating functions, conditioned sums, and stable laws is also classical; Janson [11], especially Sections 17–18, provides an extensive treatment. We include the particular Fourier proof needed for a varying feedback parameter, so that the uniformity is not left to an unstated invocation of a fixed-distribution theorem.

The exact boundary formula, the nonlinear critical charts, the order-three correction polynomials, and the explicit uniform order chart are the results established here for (1.1). Their novelty relative to the inspected repository is documented above. A focused literature search found the standard general mechanisms and their antecedents; it did not establish publication priority for every model-specific formula. No theorem is advertised as a new general principle of probability or transseries.

1.3 Conventions

Write

$$a_n(s,u) = [r^n]v(r), \qquad A_s = \zeta(s-1) \quad (s>2), \qquad U_{\mathrm{c}}(s) = \sqrt{2\zeta(s) + A_s^2}.$$

An assertion about the coefficients a_n always concerns the germ $v(0) = 0$ of (1.1). It is not automatically an assertion about the ordinary compositional inverse of $-\mathrm{Li}_s(-\mathrm{e}^x)$. The incoming source already makes this distinction; the special dilogarithm identification does not extend to arbitrary s.

All square roots on a positive real argument denote the positive root. Local inverse branches are selected by positivity for positive real variables and then continued to a slit sector. A *convergent transseries chart* means a holomorphic function of finitely many small coordinates whose ordinary multivariable Taylor series converges absolutely after substitution. This claim is stronger than a Poincaré asymptotic expansion. The asymptotic coefficient expansions obtained after singularity transfer need not themselves converge.

2 The complete radius and coefficient classification

Here and below, an equivalent $a_n \sim CR^{-n}n^{-p}(\log n)^{-q}$ has a strictly positive constant C. This is useful because it proves that the specified point really is singular, in addition to describing the coefficients.

Theorem 2.1 (Complete feedback phase diagram). *For every $s > 0$ and $u > 0$, (1.1) has a unique holomorphic germ at zero. Every coefficient $a_n(s,u)$ is positive. The following list gives its exact Taylor radius and coefficient equivalent. In every case the positive radius is the unique singularity on the circle of convergence, and the Taylor series converges absolutely on that circle.*

1. *If $0 < s \le 2$, or if $s > 2$ and $u < U_c(s)$, there is a unique $t_* \in (0,1)$ such that*
$$u^2 = 2\,\mathrm{Li}_s(t_*) + \mathrm{Li}_{s-1}(t_*)^2. \tag{2.1}$$
With $w_* = \mathrm{Li}_{s-1}(t_*)$ and $R = t_* e^{w_* - u}$,
$$a_n \sim \sqrt{\frac{w_*}{2\pi(1+\mathrm{Li}_{s-2}(t_*))}}\, R^{-n} n^{-3/2}. \tag{2.2}$$
This is the interior theorem of [1].

2. *Suppose $s > 2$ and $u > U_c(s)$. Define*
$$A = \zeta(s-1), \quad b = \sqrt{u^2 - 2\zeta(s)}, \quad v_b = u - b, \quad R_b = e^{-v_b}. \tag{2.3}$$
Then $b > A$, the radius is R_b, and
$$\boxed{a_n \sim \frac{b^{s-1}}{(b-A)^s}\, R_b^{-n} n^{-s}.} \tag{2.4}$$
This proves the candidate in the incoming source's first research question, including every integral $s \ge 3$.

3. *Suppose $u = U_c(s)$, $s > 2$, so that $b = A$ in (2.3). The radius is again R_b. If $2 < s < 3$, put $\alpha = s - 1$ and $\rho = 1/\alpha$. Then*
$$a_n \sim \frac{(A/\Gamma(-\alpha))^{1/\alpha}}{-\Gamma(-1/\alpha)}\, R_b^{-n} n^{-1-1/\alpha}. \tag{2.5}$$
At $s = 3$, with $A = \zeta(2)$,
$$a_n \sim \sqrt{\frac{A}{\pi}}\, R_b^{-n} n^{-3/2} (\log n)^{-1/2}. \tag{2.6}$$
If $s > 3$, then
$$a_n \sim \sqrt{\frac{A}{2\pi(1+\zeta(s-2))}}\, R_b^{-n} n^{-3/2}. \tag{2.7}$$

The transition has two independent parameters. Changing u determines whether the first obstruction is an interior fold or the endpoint $t = 1$. Once the endpoint and the fold coincide, changing s determines the regularity of the endpoint. In particular, the vanishing limit of the finite-curvature constant as $s \downarrow 3$ does not describe the $s = 3$ law: the logarithm changes the scale.

2.1 Positivity and the positive real branch

Lemma 2.2 (Positive germ and parametrization). *The unique germ of (1.1) has $a_n \ge 1/(un^s) > 0$. On the increasing part of the positive real branch it is parametrized by*
$$f_u(t) = u - \sqrt{u^2 - 2\,\mathrm{Li}_s(t)}, \qquad v = f_u(t), \qquad r = t e^{-f_u(t)}. \tag{2.8}$$
The first stationary point of $r(t)$, when it exists, is exactly (2.1).

Proof. Write

$$\Phi(r,v)=\frac{v^2}{2u}+\frac{1}{u}\sum_{n\ge1}\frac{r^n e^{nv}}{n^s}.$$

The implicit function theorem applies to $v=\Phi(r,v)$ at $(0,0)$, because $\Phi_v(0,0)=0$. Starting with zero and iterating Φ formally produces nonnegative coefficients. If two zero-constant-term series agree through degree $N-1$, their images agree through degree N, so every coefficient stabilizes after finitely many iterations. The direct term $r^n/(un^s)$ proves the strict lower bound.

Solving the quadratic at $v=0, t=0$ selects (2.8). Where its square root is positive,

$$\frac{\mathrm{d}}{\mathrm{d}t}\log r(t)=\frac{1}{t}\left(1-\frac{\mathrm{Li}_{s-1}(t)}{\sqrt{u^2-2\,\mathrm{Li}_s(t)}}\right). \tag{2.9}$$

The function $2\,\mathrm{Li}_s(t)+\mathrm{Li}_{s-1}(t)^2$ is strictly increasing on $(0,1)$. Its limit is infinite for $s\le2$ and equals $U_c(s)^2$ for $s>2$. At an interior stationary point the radicand is $\mathrm{Li}_{s-1}(t)^2>0$, so the stationary point occurs before a possible zero of the radicand. These facts prove all assertions. □

2.2 Global convergence and absence of competing singularities

We supply the details for $u\ge U_c(s)$, the previously unproved range. The interior proof uses the same argument with t_* in place of 1.

For $u\ge U_c(s)$ the radicand is positive on $[0,1]$, and (2.9) is positive for $0<t<1$. The parametrized solution therefore increases from $(0,0)$ to (R_b,v_b). Fix $0\le r<R_b$. The real iterates of $\Phi(r,\cdot)$ starting at zero are bounded above by this solution. Monotone convergence, first in the iteration and then in the coefficient index, shows that the formal series converges at this r. Thus its radius is at least R_b. Its sum agrees with the parametrized branch by local uniqueness and continuation. Taking $r\uparrow R_b$ gives

$$\sum_{n\ge1}a_nR_b^n=v_b<\infty. \tag{2.10}$$

Consequently the series converges absolutely and uniformly throughout the closed disk $|r|\le R_b$.

Let $|r|=R_b$, $r\ne R_b$. Strict positivity of every a_n gives $|v(r)|<v_b$: equality in the triangle inequality would force r^n/R_b^n to have the same argument for every positive integer n. It follows that

$$|re^{v(r)}|<R_be^{v_b}=1.$$

The defining equation is analytic at this boundary point, and

$$|\Phi_v(r,v(r))|\le\frac{|v(r)|+\mathrm{Li}_{s-1}(|re^{v(r)}|)}{u}<\frac{v_b+A}{u}\le1. \tag{2.11}$$

The implicit function theorem extends the germ across every such point. Therefore R_b is the only possible singularity on the circle. The local singular expansions below prove that it is indeed singular and hence that the radius is exactly R_b.

2.3 An exact integral normal form

Define, for $s > 2$ and x in a small slit neighborhood of zero,

$$D_s(x) = \zeta(s) - \mathrm{Li}_s(\mathrm{e}^{-x}), \qquad K_s(x) = Ax + \frac{x^2}{2} - D_s(x). \tag{2.12}$$

Since $D_s'(x) = \mathrm{Li}_{s-1}(\mathrm{e}^{-x})$,

$$\boxed{K_s(x) = \int_0^x (x-t)\big(1 + \mathrm{Li}_{s-2}(\mathrm{e}^{-t})\big)\,\mathrm{d}t.} \tag{2.13}$$

The integral is along the radial segment on the chosen branch. It converges at zero for $s > 2$, including $2 < s \le 3$. Indeed the singular part of its integrand is of order t^{s-3} or $\log t$, and $s - 3 > -1$.

Put $x = -\operatorname{Log} t$ and $d = -\operatorname{Log}(r/R_\mathrm{b})$. The parametrization gives the exact equations

$$d = \Psi_b(x) := x + b - \sqrt{b^2 + 2D_s(x)}, \tag{2.14}$$

$$v(r) = v_\mathrm{b} + d - x, \tag{2.15}$$

$$(b + x)d - \frac{d^2}{2} = (b - A)x + K_s(x). \tag{2.16}$$

The last identity follows by squaring the first one, using the root that tends to b. Conversely, near zero, that root condition recovers (2.14); no extra local inverse branch is introduced by the squaring. At the critical parameter the linear term disappears:

$$\boxed{K_s(x) = (A + x)d - \frac{d^2}{2}.} \tag{2.17}$$

This single equation is the starting point of all the convergent critical charts below.

2.4 The local singularities

For nonintegral $s > 2$, set $\alpha = s - 1$. Equation (1.3) gives the convergent identity

$$K_s(x) = \Gamma(-\alpha)x^\alpha + x^2 B_s(x), \qquad B_s(x) = \frac{1 + \zeta(s-2)}{2} + \sum_{k\ge 3} \zeta(s-k)\frac{(-1)^k x^{k-2}}{k!}. \tag{2.18}$$

For an integer $m \ge 3$ the corresponding resonant formula is

$$K_m(x) = \frac{x^2}{2} + \sum_{\substack{k\ge 2\\ k\ne m-1}} \zeta(m-k)\frac{(-x)^k}{k!} + \frac{(-x)^{m-1}}{(m-1)!}\big(H_{m-1} - \operatorname{Log} x\big). \tag{2.19}$$

The separate $x^2/2$ remains present when $m = 3$. In that case

$$K_3(x) = \frac{x^2}{2}\left(\log\frac{1}{x} + \frac{5}{2}\right) + \frac{x^3}{12} - \frac{x^4}{288} + O(x^6). \tag{2.20}$$

The absence of the x^5 term is $\zeta(-2) = 0$.

At $b = A$, (2.17) now gives

$$\Psi_A(x) \sim \begin{cases} \dfrac{\Gamma(1-s)}{A} x^{s-1}, & 2 < s < 3, \\ \dfrac{x^2}{2A} \log(1/x), & s = 3, \\ \dfrac{1+\zeta(s-2)}{2A} x^2, & s > 3. \end{cases} \tag{2.21}$$

All equivalents hold uniformly on closed subsectors of the slit. For $b > A$ the leading term is instead $\Psi_b(x) \sim \kappa x$, where $\kappa = 1 - A/b > 0$.

Lemma 2.3 (Sectorial inverse and dented continuation)**.** *Each boundary branch above has a holomorphic inverse $x(d)$ in a sufficiently small sector $|\arg d| < \pi - \eta$, for any fixed $0 < \eta < \pi$, selected by $x > 0$ when $d > 0$. Together with (2.11), this supplies a dented disk at $r = R_{\mathrm{b}}$ of the type required for singularity transfer.*

Proof. For $\Psi(x) \sim cx^p$, $c > 0$, $p \geq 1$, take $x_0 = (d/c)^{1/p}$. On a small fixed relative circle $|x/x_0 - 1| = \epsilon$, the normalized function $\Psi(x)/d - 1$ converges uniformly to $(x/x_0)^p - 1$, which has one simple zero inside. Rouché's theorem gives a unique root near x_0. The differentiated convergent expansions show that the derivative is nonzero there. The local implicit inverses glue by uniqueness. For $\Psi(x) \sim cx^2 \log(1/x)$, the same proof uses $x_0 = (2d/(c \log(1/d)))^{1/2}$; the slowly varying logarithm has ratio tending to one on the relative circle. These arguments use the differentiated expansions, rather than differentiating an arbitrary unspecified $o(1)$.

The variable $d = -\operatorname{Log}(r/R_{\mathrm{b}})$ is analytic near $r = R_{\mathrm{b}}$ and equals $1 - r/R_{\mathrm{b}} + O((1 - r/R_{\mathrm{b}})^2)$. Its slit sector supplies the local dent. The rest of the circle has analytic extensions by (2.11). A finite cover of its compact arcs completes a dented disk after its thickness is decreased. □

2.5 Proof of the coefficient laws

For nonintegral $s > 2$ and $b > A$, regard x and x^α as independent local coordinates. In (2.14), the coefficient of its first nonanalytic term is $\Gamma(-\alpha)/b$. The inverse therefore has first nonanalytic term

$$-\frac{\Gamma(-\alpha)}{b\kappa^{\alpha+1}} d^\alpha.$$

One way to verify the factor is to substitute the analytic inverse $x_{\mathrm{an}}(d) = d/\kappa + O(d^2)$ and divide the perturbation by $\Psi'_{\mathrm{an}}(0) = \kappa$. Equation (2.15) changes its sign. Thus the singular term of v is

$$\frac{\Gamma(1-s)}{b\kappa^s} d^{s-1}. \tag{2.22}$$

Transfer of $(1-z)^{s-1}$ divides by $\Gamma(1-s)$ and gives (2.4).

The integer case can be checked directly, avoiding any interchange of a parameter limit and a coefficient asymptotic. For $s = m \geq 3$, put $j = m - 1$. The singular term of v is

$$\frac{(-1)^{j+1}}{b\, j!\, \kappa^m} d^j \operatorname{Log} d. \tag{2.23}$$

The exact logarithmic transfer equivalent is

$$[z^n](1-z)^j \operatorname{Log}(1-z) \sim (-1)^{j+1} j!\, n^{-j-1}.$$

The signs and factorials cancel, giving the same positive constant $1/(b\kappa^m)$.

At criticality, inversion of (2.21) and (2.15) yields respectively

$$v = v_{\mathrm{b}} - (A/\Gamma(-\alpha))^{1/\alpha} d^{1/\alpha} + o(d^{1/\alpha}), \qquad 1 < \alpha < 2,$$
$$v = v_{\mathrm{b}} - 2\sqrt{A}\, d^{1/2} (\log(1/d))^{-1/2} \left(1 + O\left(\frac{\log\log(1/d)}{\log(1/d)}\right)\right), \qquad s = 3,$$
$$v = v_{\mathrm{b}} - \sqrt{\frac{2A}{1+\zeta(s-2)}}\, d^{1/2} + o(d^{1/2}), \qquad s > 3.$$

The term d in (2.15) is locally analytic and is smaller than the displayed singular terms. Apply Lemma 2.3 and the power–logarithm transfer theorem [10]. Since $\Gamma(-1/2) = -2\sqrt{\pi}$, this proves (2.5)–(2.7). The local expansions prove genuine singularity and complete the radius argument. At an interior point the usual implicit quadratic expansion has amplitude $\sqrt{2w_*/(1+\mathrm{Li}_{s-2}(t_*))}$; the same transfer gives (2.2). This also reproves the part credited to the incoming report.

Corollary 2.4 (The physical convergence threshold for every order)**.** *For every $s > 0$ there is a unique $t_s \in (0,1)$ satisfying*

$$\log t_s + \mathrm{Li}_{s-1}(t_s) = 0. \tag{2.24}$$

Put $u_s^2 = 2\,\mathrm{Li}_s(t_s) + \mathrm{Li}_{s-1}(t_s)^2$. The Taylor series of v evaluated at $r = -\mathrm{e}^{-u}$ converges if and only if $u \geq u_s$. At equality it converges absolutely.

Proof. The left side of (2.24) is strictly increasing from $-\infty$ to a positive limit. In the interior regime, $\mathrm{e}^{-u}/R = \mathrm{e}^{-w_*}/t_*$, so the condition that the evaluation point lie on or within the convergence circle is precisely $\log t_* + w_* \geq 0$. Monotonicity of the characteristic equation proves the stated threshold. If $s > 2$, then $u_s < U_{\mathrm{c}}(s)$. For all $u \geq U_{\mathrm{c}}(s)$ the boundary radius instead satisfies $\mathrm{e}^{-u}/R_{\mathrm{b}} = \mathrm{e}^{-b} < 1$. Absolute convergence at equality follows from Theorem 2.1. □

The incoming report proves this last statement for $0 < s \leq 2$. The boundary classification removes that restriction.

3 Convergent inverse charts at the critical endpoint

We now resolve the critical singularity beyond its leading equivalent. Throughout this section $u = U_{\mathrm{c}}(s)$, $s > 2$, and the inverse is selected by $x > 0$ when $d > 0$. Its exact equation is (2.17). All fractional powers are continued from the positive axis in the sector supplied by Lemma 2.3. A power of a function near 1 uses the Taylor-series logarithm there. Every chart below is holomorphic in finitely many independent small coordinates; its convergent Taylor series becomes an exact generalized power or logarithmic series after the indicated substitution.

3.1 The range $2 < s < 3$

Put

$$\alpha = s - 1, \qquad \beta = 2 - \alpha = 3 - s, \qquad C = \Gamma(1-s) > 0,$$

and use B_s from (2.18). Thus $K_s(x) = Cx^\alpha + x^2 B_s(x)$. Write

$$B_0 = B_s(0) = \frac{1+\zeta(s-2)}{2}, \qquad B_1 = B_s'(0) = -\frac{\zeta(s-3)}{6}.$$

The two small exponents $\alpha - 1$ and β are both positive and sum to one. This relation removes the need for a third generator.

Theorem 3.1 (A convergent chart with polynomial sectors)**.** *Define*

$$q = \left(\frac{Ad}{C}\right)^{1/\alpha}, \qquad p = q^{\alpha-1}, \qquad z = q^{\beta}. \tag{3.1}$$

There is a unique holomorphic germ $U(p,z)$ at $(0,0)$, with $U(0,0) = 1$, satisfying

$$U^{\alpha} + \frac{z}{C}U^2 B_s(pzU) = 1 + \frac{pz}{A}U - \frac{C}{2A^2}p^2 z. \tag{3.2}$$

The critical inverse is exactly

$$x = qU(q^{\alpha-1}, q^{\beta}). \tag{3.3}$$

Moreover,

$$U(p,z) = 1 + \sum_{\ell \geq 1} P_\ell(p) z^\ell, \qquad P_\ell \in \mathbb{R}[p], \qquad \deg P_\ell \leq 2\ell, \tag{3.4}$$

with normal convergence on a sufficiently small bidisk.

Proof. Substitute $x = qU$ and $d = Cq^{\alpha}/A$ into (2.17), then divide by Cq^{α}. Since $q = pz$ and $q^{\alpha} = p^2 z$, the result is (3.2). Its derivative in U at $(p, z, U) = (0, 0, 1)$ is $\alpha \neq 0$. The holomorphic implicit function theorem gives the germ. Substitution recovers the original equation, and Lemma 2.3 identifies the selected inverse.

At $z = 0$ the solution is $U = 1$. In the perturbation side obtained after isolating $U^{\alpha} - 1$, every monomial $p^j z^\ell$ has $\ell \geq 1$ and $j \leq 2\ell$. The three sources of such monomials are pz, $p^2 z$, and $p^j z^{j+1}$ from the analytic tail. Induction on the z-degree in the implicit coefficient recursion preserves this support condition. It proves the polynomial assertion; holomorphy proves convergence. □

There is a finite formula for every sector polynomial. Put

$$F_0(h) = (1+h)^{\alpha} - 1,$$
$$\Psi(p,z,h) = \frac{pz}{A}(1+h) - \frac{Cp^2 z}{2A^2} - \frac{z}{C}(1+h)^2 B_s(pz(1+h)).$$

Then

$$P_\ell(p) = \sum_{k=1}^{\ell} \frac{1}{k}[z^\ell h^{k-1}] \left\{ \frac{h}{F_0(h)} \Psi(p,z,h) \right\}^k. \tag{3.5}$$

Indeed, insert an auxiliary multiplier on Ψ in $F_0(h) = \Psi$, rewrite this as a Lagrange equation for h, and extract its coefficients. Since Ψ is divisible by z, only $k \leq \ell$ can contribute. This finite extraction agrees with the Taylor coefficients of the already established analytic germ; no convergence assertion about the auxiliary multiplier is needed.

For example,

$$P_1(p) = -\frac{B_0}{\alpha C} + \frac{p}{\alpha A} - \frac{Cp^2}{2\alpha A^2}, \tag{3.6}$$

$$P_2(p) = \frac{1}{\alpha}\left\{ \left(\frac{p}{A} - \frac{2B_0}{C}\right) P_1(p) - \frac{B_1 p}{C} - \frac{\alpha(\alpha-1)}{2} P_1(p)^2 \right\}. \tag{3.7}$$

The whole ladder at $p = 0$ has the closed form

$$P_\ell(0) = \frac{(-B_0/C)^\ell}{\alpha\, \ell!} \frac{\Gamma((2\ell+1)/\alpha)}{\Gamma(1 + (1+\ell\beta)/\alpha)}. \tag{3.8}$$

To prove it, set $p = 0$ in the chart and put $w = U^\alpha$. The equation becomes $w + (B_0/C)zw^{2/\alpha} = 1$; ordinary Lagrange inversion applied to $w^{1/\alpha}$ gives (3.8).

The terms of total Taylor degree at most two are

$$U = 1 - \frac{B_0}{\alpha C}z + \frac{1}{\alpha A}pz + \frac{(5-\alpha)B_0^2}{2\alpha^2 C^2}z^2 + O((|p|+|z|)^3). \tag{3.9}$$

Their asymptotic order after substitution depends on s; equal powers must be collected when the two exponents are rationally related. In particular, if $B_0 \neq 0$,

$$x = q - \frac{B_0}{\alpha C}q^{1+\beta} + o(q^{1+\beta}).$$

There is a genuine cancellation possibility. If $\zeta(s-2) = -1$, then $B_0 = 0$ and the entire ladder (3.8) vanishes. In this case $U(p,0) = U(0,z) = 1$, so $U - 1$ is divisible by $pz = q$. Equation (3.6) therefore gives instead

$$x = q + \frac{q^2}{\alpha A} + o(q^2). \tag{3.10}$$

No uniqueness claim about an order satisfying this cancellation condition is needed for the result.

3.2 An exact Lambert chart at order three

Let $s = 3$, $A = \zeta(2)$, and define the analytic tail C_3 by

$$K_3(x) = \frac{x^2}{2}\left(\log\frac{1}{x} + \frac{5}{2}\right) + x^3 C_3(x).$$

By (2.20), $C_3(x) = 1/12 - x/288 + O(x^3)$; its Taylor series converges for $|x| < 2\pi$.

Theorem 3.2 (Convergent marginal inverse)**.** *For small positive d, put*

$$\Lambda = -\frac{1}{2}W_{-1}(-4Ade^{-5}), \qquad \tau = \sqrt{\frac{2Ad}{\Lambda}}. \tag{3.11}$$

Thus $\Lambda = \log(1/\tau) + 5/2$ and $\tau^2\Lambda = 2Ad$. There is a holomorphic germ $U(p,w)$ with $U(0,0) = 1$ such that

$$x = \tau U(\tau\Lambda, \Lambda^{-1}). \tag{3.12}$$

Its defining equation is

$$U^2(1 - w\log U) + 2pw^2U^3C_3(pwU) = 1 + \frac{pw}{A}U - \frac{p^2w}{4A^2}. \tag{3.13}$$

The germ $U - 1$ is divisible by pw, and

$$x = \tau + \tau^2\frac{\Lambda/A - 1/6}{2\Lambda - 1} + O(\tau^3\Lambda). \tag{3.14}$$

Proof. The branch W_{-1} in (3.11), with the standard convention of [9], is selected because $\Lambda \to +\infty$. The core solves $\tau^2\{\log(1/\tau) + 5/2\} = 2Ad$ exactly. Put $p = \tau\Lambda$, $w = 1/\Lambda$, so that $\tau = pw$ and $d = p^2w/(2A)$. Substitution of $x = \tau U$ in (2.17), followed by division by Ad, gives (3.13). Its derivative in U at the origin is 2, so the holomorphic implicit function theorem applies. At $p = 0$ or $w = 0$, uniqueness gives $U = 1$, proving divisibility.

At fixed w, the coefficient of p is

$$[p]U(p,w) = \frac{w/A - w^2/6}{2-w}.$$

The remaining terms are $O(p^2 w)$ uniformly for small w. Multiplication by τ and physical substitution yield (3.14). The real inverse agrees by Lemma 2.3, and its sectorial continuation fixes the complex branch. □

Here too every coefficient is computable by finite extraction. For $U = 1 + \sum_{n\ge 1} u_n(w)p^n$, set

$$F_0(h,w) = (1+h)^2\{1 - w\log(1+h)\} - 1,$$
$$\Psi(p,w,h) = \frac{pw}{A}(1+h) - \frac{p^2 w}{4A^2} - 2pw^2(1+h)^3 C_3(pw(1+h)).$$

Then

$$u_n(w) = \sum_{k=1}^{n} \frac{1}{k}[p^n h^{k-1}] \left\{ \frac{h}{F_0(h,w)} \Psi(p,w,h) \right\}^k. \tag{3.15}$$

These are rational functions of w, with their only possible finite pole at $w = 2$, of order at most $2n-1$. Indeed, $F_0(h,w)/h = 2 - w + O(h)$; extracting h^{k-1} from its negative kth power requires at most $2k-1$ inverse factors of $2-w$. The assertion concerns coefficient functions of a chart proved near $w = 0$; it does not assert a singularity of a global physical inverse.

3.3 Noninteger and integer orders above three

For noninteger $s > 3$, put

$$\mu = s - 3 > 0, \quad C = \Gamma(1-s), \quad B = \frac{1+\zeta(s-2)}{2} > 0, \quad q = \sqrt{Ad/B}.$$

Equation (2.18) reads $K_s(x) = x^2 B_s(x) + Cx^{\mu+2}$, with $B_s(0) = B$.

Theorem 3.3 (Convergent power and logarithmic charts)**.** *For noninteger $s > 3$, the critical inverse is*

$$x = qU(q, q^\mu), \tag{3.16}$$

where U is holomorphic at $(0,0)$, $U(0,0) = 1$, and

$$U^2 \frac{B_s(qU)}{B} + \frac{C}{B} z U^{\mu+2} = 1 + \frac{q}{A}U - \frac{Bq^2}{2A^2}. \tag{3.17}$$

For an integer $m \ge 4$, let

$$\nu = m - 3, \quad A = \zeta(m-1), \quad B = \frac{1+\zeta(m-2)}{2}, \quad c_m = \frac{(-1)^{m-1}}{(m-1)!}.$$

Define the analytic function B_m by

$$K_m(x) = x^2 B_m(x) + c_m x^{\nu+2}\log(1/x), \qquad B_m(0) = B.$$

With $q = \sqrt{Ad/B}$, the integer-order inverse is

$$x = qU(q, q^\nu \log(1/q)), \tag{3.18}$$

where $U(0,0) = 1$ is holomorphic and satisfies

$$U^2 \frac{B_m(qU)}{B} + \frac{c_m}{B} z U^{\nu+2} - \frac{c_m}{B} q^\nu U^{\nu+2}\log U = 1 + \frac{q}{A}U - \frac{Bq^2}{2A^2}. \tag{3.19}$$

Both formulas are exact convergent two-generator expansions.

Proof. In each case substitute $x = qU$, $d = Bq^2/A$ into (2.17) and divide by Bq^2. For the integer case use $\log(1/(qU)) = \log(1/q) - \log U$. The respective equations follow. Their derivatives in U at the origin are 2, proving the holomorphic charts. The positive and sectorial branch identifications are the same as in Theorem 3.1. □

Every mixed coefficient has a finite Lagrange formula of the same kind as (3.5). More precisely, let $F_0(h) = (1+h)^2 - 1$ and move all remaining terms of the relevant chart equation to its right side, denoted $\Psi(q, z, h)$. Then

$$[q^n z^\ell](U-1) = \sum_{k=1}^{n+\ell} \frac{1}{k} [q^n z^\ell h^{k-1}] \left\{ \frac{h}{F_0(h)} \Psi(q, z, h) \right\}^k. \tag{3.20}$$

The constant term of Ψ in (q, z) vanishes, so the sum is finite. For the noninteger case the first bivariate terms are

$$U(q, z) = 1 - \frac{C}{2B} z + \left(\frac{1}{2A} + \frac{\zeta(s-3)}{12B} \right) q + O(|q|^2 + |qz| + |z|^2), \tag{3.21}$$

and the pure fractional sectors are

$$[z^k] U(0, z) = \frac{(-C/B)^k}{2k!} \frac{\Gamma(((\mu+2)k+1)/2)}{\Gamma((\mu k+3)/2)}. \tag{3.22}$$

The latter follows by Lagrange inversion of $U^2 + (C/B) z U^{\mu+2} = 1$.

At integer order, collecting equal powers in the absolutely convergent double series gives

$$x = q \sum_{j \geq 0} q^j Q_j(\log(1/q)), \qquad \deg Q_j \leq \lfloor j/\nu \rfloor. \tag{3.23}$$

This bound is attained at every $j = \nu k$: the highest logarithmic coefficient comes from the q-constant part of the chart and equals

$$[L^k] Q_{\nu k}(L) = \frac{(-c_m/B)^k}{2k!} \frac{\Gamma(((\nu+2)k+1)/2)}{\Gamma((\nu k+3)/2)} \neq 0. \tag{3.24}$$

In particular, order four requires the resonant constant $B_4'(0) = -H_3/6 = -11/36$, and yields

$$x = q + \frac{q^2}{12B} \log \frac{1}{q} + \left(\frac{1}{2A} + \frac{11}{72B} \right) q^2 + O(q^3 \log^2(1/q)). \tag{3.25}$$

The harmonic constant is essential: directly substituting $s = 4$ in the noninteger coefficient (3.21) would encounter $\zeta(1)$ and lose the resonant cancellation.

4 Every logarithmic coefficient correction at order three

In this section $s = 3$, $u = U_c(3)$, $A = \zeta(2)$, and $a_n = a_n(3, U_c(3))$. The exact Lambert core from Theorem 3.2 determines all inverse powers of the large logarithm in the leading singular sector. The analytic feedback correction satisfies $x - \tau = O(\tau^2)$ and is smaller than every such logarithmic correction relative to τ.

4.1 A convergent logarithmic expansion of the exact core

Put

$$L = \log(1/d), \qquad c = 5 - \log(4A), \qquad T = 2\Lambda. \tag{4.1}$$

The identities defining Λ are equivalent to

$$T - \log T = L + c, \qquad \tau = 2\sqrt{Ad/T}. \tag{4.2}$$

Both the factor $4A$ and the additive constant 5 enter the final coefficient correction.

Proposition 4.1 (All core logarithmic polynomials)**.** *For sufficiently small positive d,*

$$\tau = 2\sqrt{Ad}\, L^{-1/2} \sum_{j\ge 0} \frac{P_j(\log L + c)}{L^j}, \tag{4.3}$$

where this series converges, and

$$P_0(Y) = 1, \qquad P_j(Y) = -\frac{1}{2j}[h^{j-1}](1+h)^{-3/2}\{Y + \log(1+h)\}^j \quad (j \ge 1). \tag{4.4}$$

Each P_j has degree j. For every fixed $M \ge 0$, truncation after $j = M$ has relative remainder $O((\log L)^{M+1}/L^{M+1})$.

Proof. Write $T = L(1+V)$. Equation (4.2) becomes

$$V = \eta + p\log(1+V), \qquad p = L^{-1}, \quad \eta = (\log L + c)/L.$$

The implicit derivative in V at $(p,\eta,V) = (0,0,0)$ is one. Thus V and $(1+V)^{-1/2}$ are holomorphic in the two small coordinates (p,η). After setting $\eta = pY$, apply ordinary Lagrange inversion to $V = p\{Y + \log(1+V)\}$ and the function $(1+V)^{-1/2}$. It gives (4.4). Grouping the normally convergent double series by total degree gives (4.3); both p and pY are small on the physical curve. The leading coefficient of P_j is $(-1)^j\binom{2j}{j}/4^j$, which is nonzero. Cauchy bounds for the double series give the stated remainder and its uniform counterpart on closed subsectors. □

The first polynomials are

$$P_1(Y) = -\frac{Y}{2}, \tag{4.5}$$

$$P_2(Y) = \frac{3Y^2}{8} - \frac{Y}{2}, \tag{4.6}$$

$$P_3(Y) = -\frac{5Y^3}{16} + Y^2 - \frac{Y}{2}. \tag{4.7}$$

In particular,

$$x = 2\sqrt{Ad/L}\left[1 - \frac{\log L - \log(4A) + 5}{2L} + O\left(\frac{(\log L)^2}{L^2}\right)\right]. \tag{4.8}$$

The $O(\tau^2)$ correction from (3.14) is absorbed here because it is smaller than every fixed inverse power of L relative to the leading term.

4.2 A finite formula after singularity transfer

It is useful to state the transfer calculation explicitly: otherwise the shift from the Gamma factor can change the constant in the answer. For a near $1/2$ define

$$G(a) = \frac{1}{\Gamma(-a)}, \qquad M_k(a) = \frac{(-1)^k G^{(k)}(a)}{k!\,G(a)}. \tag{4.9}$$

For the finite sums of functions $L^{-\beta}P(\log L)$ used here, power–logarithm transfer [10] can be expressed as the asymptotic operator

$$[z^n](1-z)^a F\left(\log\frac{1}{1-z}\right) \sim \frac{n^{-a-1}}{\Gamma(-a)}\,\frac{\Gamma(-a)}{\Gamma(-a+\partial_L)}F(L)\Big|_{L=\log n}, \tag{4.10}$$

where the operator means $\sum_{k\ge 0} M_k(a)\partial_L^k$ as an asymptotic derivative expansion. No convergence of this infinite operator is claimed. For a finite target order only finitely many derivatives contribute.

The signs can be checked directly on the Hankel contour used in singularity analysis. The substitution $1-z = t/n$ replaces the logarithmic argument by $\log n - \log t$. Taylor expansion of F there contributes $(-1)^k F^{(k)}(\log n)(\log t)^k/k!$. Its Hankel integral is $G^{(k)}(a)$, giving (4.10). The analytic continuation from Lemma 2.3 permits the contour deformation. Truncating the logarithmic expansion and its differentiated versions gives the usual uniform remainder estimates; powers of $1/n$ from the contour scaling are smaller than every fixed inverse power of $\log n$.

Theorem 4.2 (All logarithmic coefficient corrections)**.** *For $N \ge 0$ define the polynomial*

$$\mathcal{Q}_N(Y) = \sum_{j=0}^{N} M_{N-j}(1/2) \prod_{\ell=0}^{N-j-1}\left(\frac{\mathrm{d}}{\mathrm{d}Y} - j - \frac{1}{2} - \ell\right) P_j(Y), \tag{4.11}$$

where the empty product is one. For every fixed $M \ge 0$,

$$R_{\mathrm{b}}^n a_n = \sqrt{\frac{A}{\pi}}\, n^{-3/2}(\log n)^{-1/2}\left[\sum_{N=0}^{M}\frac{\mathcal{Q}_N(\log\log n + c)}{(\log n)^N} + O\left(\frac{(\log\log n)^{M+1}}{(\log n)^{M+1}}\right)\right]. \tag{4.12}$$

Thus every correction is supplied by finite polynomial and Gamma-derivative operations.

Proof. For the singular part of $v = v_{\mathrm{b}} + d - x$, replace x by the exact core τ. The omitted terms $x - \tau = O(\tau^2)$ contribute at coefficient orders smaller by a positive power of n, up to powers of its logarithm. Locally analytic terms do not affect the singular expansion. Also $d = -\operatorname{Log}(r/R_{\mathrm{b}}) = 1 - r/R_{\mathrm{b}} + O((1-r/R_{\mathrm{b}})^2)$; this replacement changes only corrections smaller by a power of n.

Apply (4.10), with $a = 1/2$, to each term $L^{-j-1/2}P_j(\log L + c)$ in (4.3). Repeated differentiation gives exactly

$$\partial_L^k\{L^{-j-1/2}P_j(\log L + c)\} = L^{-j-k-1/2}\prod_{\ell=0}^{k-1}\left(\frac{\mathrm{d}}{\mathrm{d}Y} - j - \frac{1}{2} - \ell\right) P_j(Y)\Big|_{Y=\log L + c}.$$

Collecting $j + k = N$ proves (4.11). The amplitude of $-\tau$ is $-2\sqrt{A}$, and $\Gamma(-1/2) = -2\sqrt{\pi}$, giving the positive prefactor in (4.12). The finite-order transfer remainder and Proposition 4.1 prove the stated error. □

4.3 The first three terms in closed form

Let $\psi = \Gamma'/\Gamma$, $\psi_1 = \psi'$, and abbreviate their values at $-1/2$ by ψ_* and $\psi_{1,*}$. Formula (4.9) gives

$$M_0 = 1, \qquad M_1 = -\psi_*, \qquad M_2 = \frac{\psi_*^2 - \psi_{1,*}}{2}.$$

Substituting (4.5)–(4.6) into (4.11) yields

$$\begin{aligned}
\mathcal{Q}_0(Y) &= 1,\\
\mathcal{Q}_1(Y) &= -\frac{Y - \psi_*}{2},\\
\mathcal{Q}_2(Y) &= \frac{3}{8}(Y - \psi_*)^2 - \frac{1}{2}(Y - \psi_*) - \frac{3}{8}\psi_{1,*}.
\end{aligned}$$

The standard digamma recurrence and half-integer values give

$$\psi_* = 2 - \gamma - 2\log 2, \qquad \psi_{1,*} = \frac{\pi^2}{2} + 4.$$

Consequently, with

$$H_n = \log\log n - \log A + \gamma + 3, \tag{4.13}$$

we obtain the explicit refinement

$$\boxed{\begin{aligned} R_{\mathrm{b}}^n a_n = \sqrt{\frac{A}{\pi}}\, n^{-3/2} (\log n)^{-1/2} \Bigg[1 - \frac{H_n}{2\log n} \\ + \frac{\frac{3}{8}H_n^2 - \frac{1}{2}H_n - \frac{3\pi^2}{16} - \frac{3}{2}}{(\log n)^2} + O\left(\frac{(\log\log n)^3}{(\log n)^3}\right) \Bigg]. \end{aligned}} \tag{4.14}$$

The cancellation of $\log 4$ between the exact core and the digamma shift leaves the simple constant $-\log A + \gamma + 3$. The core and feedback charts are convergent analytic constructions; (4.12) and (4.14) are coefficient asymptotic expansions. These are distinct convergence statements.

5 A convergent joint crossover at order three

The fixed-order laws change at $s = 3$, while the linear term in (2.16) changes at $b = \zeta(s-1)$. We now resolve both transitions in one convergent chart. Throughout this section set

$$A(\varepsilon) = \zeta(2+\varepsilon), \quad A_0 = \zeta(2), \quad E_2(\varepsilon) = \frac{\Gamma(1-\varepsilon)}{(1+\varepsilon)(1+\varepsilon/2)}, \quad Z(\varepsilon) = \zeta(1+\varepsilon) - \varepsilon^{-1}. \tag{5.1}$$

The last two functions are holomorphic near zero, with $E_2(0) = 1$, $E_2'(0) = \gamma - 3/2$, and $Z(0) = \gamma$. Pairing the resonant terms in (1.3) gives

$$2K_{3+\varepsilon}(x) = x^2\{H(\varepsilon, x) + xT(\varepsilon, x)\}, \tag{5.2}$$

$$H(\varepsilon, x) = 1 + \zeta(1+\varepsilon) - \frac{E_2(\varepsilon)x^\varepsilon}{\varepsilon}, \tag{5.3}$$

$$T(\varepsilon, x) = 2\sum_{k\geq 3} \frac{(-1)^k \zeta(3+\varepsilon-k)}{k!} x^{k-3}. \tag{5.4}$$

Here T is jointly holomorphic for small ε and $|x| < 2\pi$; the paired Jonquière expansion converges normally on compact subsets. In particular,

$$H(0,x) = \log(1/x) + \frac{5}{2}, \qquad T(\varepsilon,0) = -\frac{\zeta(\varepsilon)}{3}, \qquad T(0,0) = \frac{1}{6}. \tag{5.5}$$

Thus (2.16) becomes the exact equation

$$d(2b + 2x - d) = 2\{b - A(\varepsilon)\}x + x^2\{H(\varepsilon,x) + xT(\varepsilon,x)\}. \tag{5.6}$$

The extra 1 in (5.3) records the quadratic feedback term; it changes the resonant constant from $3/2$ to $5/2$.

5.1 The joint order and feedback chart

Put

$$X = \frac{1}{2}\log\frac{1}{d}, \quad h = X^{-1}, \quad w = X^{-1}\log X, \quad j = \sqrt{dX}, \quad \varepsilon = \lambda h, \quad b = A(\varepsilon) + \eta j. \tag{5.7}$$

In particular $d = hj^2$. The scaled displacement $\eta = (b - A)/\sqrt{dX}$ measures when the linear and resonant quadratic terms compete. Write

$$\varphi(\lambda) = \frac{1 - \mathrm{e}^{-\lambda}}{\lambda}, \quad \varphi(0) = 1, \qquad S = \sqrt{\eta^2 + 2A_0\varphi(\lambda)}, \quad V_0 = \frac{2A_0}{S + \eta}. \tag{5.8}$$

For real λ, η, these quantities are positive. Define

$$G(\lambda,h,w,V) = \frac{1 - \mathrm{e}^{-\lambda}E_2(\lambda h)\mathrm{e}^{-\lambda w/2}V^{\lambda h}}{\lambda} + h\{1 + Z(\lambda h)\}, \tag{5.9}$$

using the logarithm near $V_0 > 0$. Its quotient is holomorphic at $\lambda = 0$, where

$$G(0,h,w,V) = 1 + \frac{w}{2} + h\left(\frac{5}{2} - \log V\right). \tag{5.10}$$

Theorem 5.1 (Convergent joint crossover)**.** *For every compact rectangle of real parameters (λ,η), there is a function $V(\lambda,\eta,h,w,j)$ holomorphic on a common complex neighborhood of that rectangle times a polydisc about $(h,w,j) = (0,0,0)$, with $V(\lambda,\eta,0,0,0) = V_0$. For all sufficiently large real X, uniformly on the rectangle, the positive feedback inverse in (5.7) is exactly*

$$\boxed{x = \sqrt{\frac{d}{X}}\,V\left(\lambda,\eta,\frac{1}{X},\frac{\log X}{X},\sqrt{dX}\right).} \tag{5.11}$$

Its defining holomorphic equation is

$$\begin{aligned} 0 = {} & V^2G(\lambda,h,w,V) + 2\eta V + h^2jV^3T(\lambda h, hjV) \\ & - 2A(\lambda h) - 2j\eta - 2hjV + hj^2. \end{aligned} \tag{5.12}$$

Thus the Taylor series in the three independent small coordinates converges after physical substitution; it includes every inverse-logarithmic order and all exponential sectors.

Proof. At $h = w = j = 0$, the right side of (5.12) is $\varphi(\lambda)V^2 + 2\eta V - 2A_0$. Its positive root is V_0, and its derivative there is $2S \neq 0$. Joint holomorphy, the implicit function theorem, and compactness give a common polydisc; uniqueness glues the local functions. Under $x = hjV$, the physical coordinates satisfy

$$x^{\varepsilon} = \mathrm{e}^{-\lambda}\mathrm{e}^{-\lambda w/2}V^{\lambda h}, \qquad hH(\varepsilon, x) = G(\lambda, h, w, V).$$

Dividing (5.6) by $d = hj^2$ now gives (5.12) exactly. For the constructed root, $b + x - d$ is close to $A_0 > 0$, so the required square-root branch in (2.14) is preserved.

For completeness the positive-root identification remains valid when $\eta < 0$. For real $s > 2$ and $b > 0$, the ratio

$$\frac{\mathrm{Li}_{s-1}(\mathrm{e}^{-x})}{\sqrt{b^2 + 2D_s(x)}}$$

strictly decreases from A/b to zero as x increases from zero to infinity. Thus $\Psi_b'(x)$ strictly increases to 1. If $b \geq A$, then Ψ_b increases from zero. If $b < A$, it first decreases to a negative minimum and then increases to infinity. In either case every $d > 0$ has exactly one positive preimage. In the latter case this root lies before the interior fold when the branch is followed from $r = 0$, since $r < R_{\mathrm{b}}$; the endpoint reference R_{b} need not itself be the Taylor radius. The positive root furnished by the chart is therefore the physical one. Finally, the asserted convergence is the ordinary convergence of the holomorphic Taylor series, not an inference from its formal substitutions. □

Corollary 5.2 (Explicit first correction)**.** *Uniformly on compact real (λ, η) sets, let*

$$B = \mathrm{e}^{-\lambda}\left\{\frac{1}{2}\log X - \log V_0 - \gamma + \frac{3}{2}\right\} + 1 + \gamma, \qquad V_1 = \frac{2\lambda\zeta'(2) - V_0^2 B}{2S}. \tag{5.13}$$

Then

$$x = \sqrt{\frac{d}{X}}\left\{V_0 + \frac{V_1}{X} + O\left(\frac{(1+\log X)^2}{X^2}\right)\right\}. \tag{5.14}$$

More generally, the $j = 0$ part of (5.11) is a convergent series $\sum_{n\geq 0} X^{-n}P_n(\log X; \lambda, \eta)$ with $\deg P_n \leq n$. Truncation after $n = N$ has uniform error $O((1+\log X)^{N+1}X^{-N-1})$ inside the braces.

Proof. Expand $\log E_2(\varepsilon) = (\gamma - 3/2)\varepsilon + O(\varepsilon^2)$ and $A(\varepsilon) = A_0 + \zeta'(2)\varepsilon + O(\varepsilon^2)$ in (5.12), first setting $j = 0$. Its coefficient of X^{-1} is $2SV_1 + V_0^2 B - 2\lambda\zeta'(2)$. The remaining assertions follow by grouping the absolutely convergent Taylor series in (h, w) by total degree. A monomial $h^i w^j$ contributes $X^{-i-j}(\log X)^j$, which proves both the degree bound and the Taylor remainder. Terms containing the third coordinate $\sqrt{dX}$ are smaller than every fixed inverse power of X and are retained in the full chart. □

At $\lambda = \eta = 0$, this recovers the explicit resonance constant:

$$x = \sqrt{\frac{2\zeta(2)d}{X}}\left\{1 - \frac{\log X - \log(2\zeta(2)) + 5}{4X} + O\left(\frac{(1+\log X)^2}{X^2}\right)\right\}. \tag{5.15}$$

The compact-parameter hypothesis is essential to the stated uniform errors; it does not by itself permit λ or η to tend to infinity with X.

5.2 Critical support and the first exponential correction

At $\eta = 0$, different analytic coordinates reveal an additional restriction on the support. Set $q = X^{-1/2}$ and $z = \sqrt{d}$, so $x = zqU$. There is a holomorphic chart $U(\lambda, q, w, z)$ through $U_0 = \sqrt{2A_0/\varphi(\lambda)}$, determined by

$$\begin{aligned} 0 = {} & U^2 G(\lambda, q^2, w, U) + zq^3 U^3 T(\lambda q^2, zqU) \\ & - 2A(\lambda q^2) - 2zqU + z^2. \end{aligned} \tag{5.16}$$

Its existence follows directly from the same implicit-function argument, since the derivative at the base root is $2\varphi(\lambda)U_0 \neq 0$.

Proposition 5.3 (Parity and the first exponential sector)**.** *The critical chart satisfies*

$$U(\lambda, -q, w, -z) = U(\lambda, q, w, z). \tag{5.17}$$

Hence the coefficient of $q^i w^j z^k$ vanishes when $i + k$ is odd. Put $h = q^2$ and, at $z = 0$, write

$$U_* = U(\lambda, q, w, 0), \quad G_* = G(\lambda, h, w, U_*), \quad M_* = \mathrm{e}^{-\lambda} E_2(\lambda h) \mathrm{e}^{-\lambda w/2} U_*^{\lambda h}.$$

The exact first exponential coefficient is

$$\partial_z U(\lambda, q, w, 0) = q \, \frac{2 + hU_*^2 \zeta(\lambda h)/3}{2G_* - hM_*}. \tag{5.18}$$

Consequently, for $x_* = \sqrt{d/X}\, U_*$ on the physical curve,

$$x = x_* + \frac{d}{X} \frac{2 + X^{-1} U_*^2 \zeta(\lambda/X)/3}{2G_* - X^{-1} M_*} + O\left(d^{3/2} X^{-1/2}\right). \tag{5.19}$$

The correction begins with $d/(X\varphi(\lambda))$. All statements are uniform for λ in compact real sets.

Proof. The defining equation (5.16) is invariant under $(q, z) \mapsto (-q, -z)$; uniqueness proves (5.17). At $z = 0$, its U derivative equals $U_*(2G_* - hM_*)$, because $G_U = -hM_*/U_*$ there. Its z derivative equals $-q^3 U_*^3 \zeta(\lambda h)/3 - 2qU_*$, by (5.5). Implicit differentiation proves (5.18). Taylor's theorem in z, followed by multiplication by zq, gives (5.19). The parity statement also independently explains why the $z = 0$ sector contains integer powers of X^{-1}. □

5.3 A local complex fold beyond a profile zero

The real crossover profile is positive, but its nonzero complex zeros are $\lambda_0 = 2\pi i k$, $k \neq 0$. The following theorem concerns local solutions of the continued normal form at $b = A$. It makes no assertion that those solutions continue the Taylor germ $v(0) = 0$ around any specified global path.

Theorem 5.4 (Local shifted complex fold)**.** *Fix $\lambda_0 = 2\pi i k \neq 0$, and put $h = 1/X$, $z = \sqrt{d}$, $\lambda = \lambda_0 + \alpha h$, $\varepsilon = h(\lambda_0 + \alpha h)$, and $x = z\omega$. There are holomorphic functions $\alpha_c(h, z)$ and $\omega_c(h, z)$ near $(0, 0)$ with*

$$\omega_c(0, 0) = 2\sqrt{A_0}, \qquad \alpha_c(0, 0) = \alpha_* := \lambda_0 \left\{ \frac{1}{2} \log(4A_0) - 2 \right\}, \tag{5.20}$$

on which the local inverse equation has a double root. An analytic local coordinate converts that equation into a square equal to $(\alpha - \alpha_c(h, z))$ times a nonvanishing holomorphic factor. In particular, the scaled-order fold curve satisfies

$$\lambda_c(X) = \lambda_0 + \frac{\lambda_0 \{ \frac{1}{2} \log(4A_0) - 2 \}}{X} + O(X^{-2} + X^{-1} \mathrm{e}^{-X}). \tag{5.21}$$

At the unshifted profile zero $\alpha = 0$, there are instead two ordinary holomorphic root germs in (h, z), through the distinct positive constants

$$\omega_j = \exp\left\{\frac{5 + W_j(-4A_0 e^{-5})}{2}\right\}, \qquad j = 0, -1. \tag{5.22}$$

Thus the unshifted profile zero itself does not force a square-root inverse-logarithmic splitting.

Proof. Define, with the logarithm near the indicated positive values,

$$\widehat{H}(h, \alpha, \omega) = 1 + Z(\varepsilon) + \frac{1 - E_2(\varepsilon)e^{-\alpha h}\omega^{\varepsilon}}{\varepsilon}.$$

The numerator is divisible by h, so this expression is holomorphic at $h = 0$. Its value there is $5/2 + \alpha/\lambda_0 - \log\omega$. The exact equation is

$$\widehat{F} := \omega^2\widehat{H} + z\omega^3 T(\varepsilon, z\omega) - 2A(\varepsilon) - 2z\omega + z^2 = 0. \tag{5.23}$$

At $h = z = 0$ it reduces to

$$\omega^2\left(\frac{5}{2} + \frac{\alpha}{\lambda_0} - \log\omega\right) = 2A_0.$$

Its double positive root is exactly (5.20); there $\widehat{F}_{\omega\omega} = -2$ and $\widehat{F}_{\alpha} = 4A_0/\lambda_0 \neq 0$. Apply the implicit function theorem to $(\widehat{F}, \widehat{F}_{\omega})$ to obtain α_c, ω_c. For the normal form, first solve $\widehat{F}_{\omega} = 0$ for a stationary point depending on (h, z, α). Taylor expansion about it factors the remaining quadratic term by a holomorphic unit. Taking its holomorphic square root gives a square coordinate. The value at the stationary point has a simple zero at α_c, because its α derivative is nonzero. This proves the claimed factorization and (5.21).

At $\alpha = 0$, the limiting equation is $\omega^2(5/2 - \log\omega) = 2A_0$. Solving it gives (5.22). Since $A_0 < 2$ and $8 < e^4$, its Lambert argument lies strictly between $-1/e$ and zero. The two stated roots are therefore distinct and simple, and the ordinary implicit function theorem supplies their local charts. No exhaustion of other complex roots is claimed. □

6 Exact coefficient identities and the feedback crossover

The local inverse charts describe the function near its endpoint. There is also a useful exact identity for its ordinary Taylor coefficients. It provides an independent explanation of the three critical coefficient scales and makes the transition in u precise. The underlying Lagrange and probability mechanism is classical [11]; closely related identities already occur in the linear critical-Hahn model [2].

6.1 Finite coefficients and a compound Poisson identity

For this subsection take $s > 2$, $b > 0$, and

$$u = \sqrt{b^2 + 2\zeta(s)}, \quad A = \zeta(s-1), \quad f_b(t) = u - \sqrt{u^2 - 2\,\mathrm{Li}_s(t)}, \quad v_b = u - b, \quad R_b = e^{-v_b}.$$

The notation R_b denotes the endpoint parametrization even when $b < A$. In that case the true convergence radius is the larger interior maximum of $te^{-f_b(t)}$; it is not R_b.

Proposition 6.1 (Exact coefficient and probability identities). *Write $f_b(t) = \sum_{k\ge1} c_k t^k$. Then $c_k > 0$,*

$$c_k = \frac{1}{uk^s} + \frac{1}{2u}\sum_{j=1}^{k-1} c_j c_{k-j}, \tag{6.1}$$

and, for every $n \ge 1$,

$$\boxed{a_n = \frac{1}{n}[t^n]\exp\bigl(nf_b(t)\bigr).} \tag{6.2}$$

Let Y_b be the nonnegative integer-valued random variable with probability generating function

$$p_b(t) = \exp(f_b(t) - f_b(1)) = \exp\left(\sum_{k\ge1} c_k(t^k - 1)\right). \tag{6.3}$$

For independent copies $Y_{b,1}, \dots, Y_{b,n}$, set $S_{n,b} = \sum_{j=1}^{n} Y_{b,j}$. Then

$$R_{\mathrm{b}}^n a_n = \frac{1}{n}\mathbb{P}(S_{n,b} = n), \qquad \mu_b := \mathbb{E}Y_b = \frac{A}{b}. \tag{6.4}$$

When $s > 3$,

$$\sigma_b^2 := \operatorname{Var} Y_b = \frac{\zeta(s-2)}{b} + \frac{A^2}{b^3}. \tag{6.5}$$

Proof. The quadratic identity $uf_b - f_b^2/2 = \operatorname{Li}_s(t)$ gives (6.1) and positivity. Since $u^2 > 2\zeta(s)$, the square-root series converges normally up to $|t| = 1$, and $\sum c_k = f_b(1) = v_{\mathrm{b}} < \infty$. Writing $t = re^{f_b(t)}$, ordinary Lagrange inversion gives

$$[r^n]f_b(t(r)) = \frac{1}{n}[t^{n-1}]f_b'(t)e^{nf_b(t)} = \frac{1}{n}[t^n]e^{nf_b(t)}.$$

The second equality follows by differentiating $e^{nf_b(t)}$. This is (6.2). Equation (6.3) is a compound Poisson generating function with jump intensities c_k. Multiplying it n times gives (6.4). Finally

$$tf_b'(t) = \frac{\operatorname{Li}_{s-1}(t)}{\sqrt{u^2 - 2\operatorname{Li}_s(t)}}.$$

Taking one, and then two, logarithmic derivatives of p_b gives the stated mean and variance. □

This identity is computationally convenient. For a specified n, let $E_k = R_{\mathrm{b}}^n[t^k]e^{nf_b(t)}$. Then

$$E_0 = e^{-nv_{\mathrm{b}}}, \qquad E_k = \frac{n}{k}\sum_{j=1}^{k} jc_jE_{k-j}, \qquad R_{\mathrm{b}}^n a_n = \frac{E_n}{n}. \tag{6.6}$$

The calculation uses only positive summands. The supplied program also checks it against a coefficient recursion in the original implicit equation, so that it is not verified solely by a second implementation of the same formula.

6.2 The normalized crossover profiles

For $1 < \alpha < 2$, define

$$g_\alpha(\xi) = \frac{1}{2\pi}\int_{\mathbb{R}} \exp\bigl((-it)^\alpha - it\xi\bigr)\,dt. \tag{6.7}$$

The branch is $(-it)^\alpha = |t|^\alpha \exp(-i\,\mathrm{sgn}(t)\pi\alpha/2)$. Since $\cos(\pi\alpha/2) < 0$, this integral converges absolutely. It is the centered, spectrally positive stable density in the normalization appropriate to this problem. Put $g_{\mathrm{N}}(\xi) = (2\pi)^{-1/2}e^{-\xi^2/2}$.

Theorem 6.2 (Uniform lattice limit and coefficient crossover). *Fix $s > 2$ and a compact interval $J \subset (0, \infty)$ for b. Define*

$$B_n(b) = \begin{cases} \big(n\Gamma(1-s)/b\big)^{1/(s-1)}, & 2 < s < 3, \\ \sqrt{n \log n/(2b)}, & s = 3, \\ \sqrt{n}\,\sigma_b, & s > 3, \end{cases} \tag{6.8}$$

and let $g_s = g_{s-1}$ in the first case and $g_s = g_{\mathrm{N}}$ in the other two. Then

$$\sup_{\substack{b \in J \\ \ell \in \mathbb{Z}}} \left| B_n(b)\mathbb{P}(S_{n,b} = \ell) - g_s\left(\frac{\ell - n\mu_b}{B_n(b)}\right)\right| \longrightarrow 0. \tag{6.9}$$

In particular, if $b = b_n \in J$ is allowed to vary and

$$\xi_n = \frac{n(1 - A/b_n)}{B_n(b_n)} \tag{6.10}$$

stays in a fixed compact real set, then

$$\boxed{a_n(s, u_n) = \frac{R_{\mathrm{b}}(s, u_n)^{-n}}{nB_n(b_n)}\big(g_s(\xi_n) + o(1)\big), \qquad u_n = \sqrt{b_n^2 + 2\zeta(s)}.} \tag{6.11}$$

The error is uniform in such a window. Both sides of the feedback threshold are included. These hypotheses force $b_n \to A$, since $B_n(b)/n \to 0$ uniformly on J.

Proof. We give a direct Fourier proof, retaining uniformity in b. Set $x = -\mathrm{i}\theta$ in the local expansion of $f_b(\mathrm{e}^{-x})$. Uniformly for $b \in J$ as $x \to 0$ on this axis,

$$f_b(\mathrm{e}^{-x}) - f_b(1) = -\mu_b x + \frac{\Gamma(1-s)}{b}x^{s-1} + O_J(x^2), \qquad 2 < s < 3, \tag{6.12}$$

$$= -\mu_b x + \frac{x^2}{2b}\log(1/x) + O_J(x^2), \qquad s = 3, \tag{6.13}$$

$$= -\mu_b x + \frac{\sigma_b^2}{2}x^2 + o_J(x^2), \qquad s > 3. \tag{6.14}$$

These follow by expanding the square root about the uniformly positive b; the coefficients of the convergent polylogarithm expansion are bounded on the compact parameter interval.

The centered characteristic function of $S_{n,b}/B_n(b)$ therefore converges, uniformly for $b \in J$ and bounded Fourier argument t, to $\exp((-\mathrm{i}t)^{s-1})$ if $2 < s < 3$, and to $\exp(-t^2/2)$ otherwise. At $s = 3$, the factor is correct because

$$\frac{n}{2bB_n(b)^2}\log B_n(b) \longrightarrow \frac{1}{2}$$

uniformly in b.

We next justify the needed L^1 convergence. The coefficients of p_b include strictly positive constant and first coefficients, so $|p_b(\mathrm{e}^{\mathrm{i}\theta})| < 1$ for $0 < |\theta| \le \pi$. Continuity and compactness give a common upper bound less than one away from a small neighborhood of zero. Its Fourier contribution is exponentially small, even after multiplication by B_n.

Near zero, taking real parts in (6.12)–(6.14) gives uniform bounds

$$|p_b(\mathrm{e}^{\mathrm{i}\theta})| \le \begin{cases} \exp(-c|\theta|^{s-1}), & 2 < s < 3, \\ \exp(-c\theta^2\log(1/|\theta|)), & s = 3, \\ \exp(-c\theta^2), & s > 3, \end{cases}$$

for a common $c > 0$ and sufficiently small $|\theta|$. In the first and third cases the substitution $\theta = t/B_n$ gives an integrable bound independent of n and b. At $s = 3$, split at $|t| = B_n^{1/2}$. Below that point, $\log(B_n/|t|)$ is a fixed positive fraction of $\log n$, and the bound is Gaussian in t. Between $B_n^{1/2}$ and a small fixed multiple of B_n, the exponent is at most $-c' n B_n^{-1} \log B_n$, after decreasing the neighborhood so that $\theta^2 \log(1/\theta)$ is increasing. Its integral tends to zero exponentially relative to the length of the interval. This proves uniform L^1 convergence in every case.

Lattice Fourier inversion, with $\theta = t/B_n$, now gives (6.9). The bound on the difference of the Fourier integrals is independent of their centering factor $\exp(-\mathrm{i}t(\ell - n\mu_b)/B_n)$, so it is uniform over all integer ℓ. Apply (6.4) with $\ell = n$ to obtain (6.11). □

6.3 Interpretation and limits of the uniform statement

At $b = A$ the profile is evaluated at zero. Direct evaluation of (6.7) at zero, followed by Euler's reflection formula, gives

$$g_\alpha(0) = \frac{\Gamma(1/\alpha)\sin(\pi/\alpha)}{\pi\alpha} = \frac{1}{-\Gamma(-1/\alpha)}. \tag{6.15}$$

The three central values in Theorem 6.2 therefore reproduce exactly the constants in Theorem 2.1. This is an independent check of the singularity-analysis calculation.

The width of the feedback transition follows from $1 - A/b = (b - A)/A + O((b - A)^2)$. In terms of $b - A$ it is

$$\begin{cases} n^{-(s-2)/(s-1)}, & 2 < s < 3, \\ \sqrt{\log n / n}, & s = 3, \\ n^{-1/2}, & s > 3. \end{cases} \tag{6.16}$$

The corresponding scale for $u - U_c$ differs by the nonzero factor A/U_c. The square-root/logarithm competition in the local inverse has thus become a Gaussian normalization with an additional $\sqrt{\log n}$ in the coefficient problem.

When $b < A$, the coefficient identity remains exact even though R_b is smaller than the actual radius: the probability in (6.4) is a lower-tail probability and contains the missing exponential factor. Confusing these two radii would give an incorrect interpretation of the crossover on that side.

The local limit theorem has a uniform *absolute* error after normalization. A compact ξ window yields the coefficient crossover stated above. It does not by itself give a uniform relative estimate for $\xi \to +\infty$ or $\xi \to -\infty$. Accordingly the fixed supercritical law was proved separately by boundary continuation and singularity transfer. A joint relative matching theorem in those expanding windows remains a further question.

7 Verification, proposed source updates, and further research

7.1 Independent algebraic and numerical checks

The arguments in this article are analytic proofs. The accompanying programs serve a separate purpose: they test constants, signs, normalizations, branch choices, and the practical sizes of the asymptotic errors. They are neither interval certificates nor formal proofs in a proof assistant. The package records the programs, their input grids, numerical precision, and complete outputs.

The finite-coefficient program computes the same quantities in three ways. For $s = 3, u = 4$, it compares the first eight coefficients from (6.2) with exact rational arithmetic in the original implicit equation. All eight agree exactly. For a larger critical case it compares the positive recurrence (6.6) at $n = 64$ with an independent 75-digit evaluation; their relative difference in the recorded environment is $1.22 \cdot 10^{-18}$. It then computes the phase tests at $n = 128, 512, 2048$ without subtracting nearly equal coefficient contributions.

Table 1: Computed coefficient divided by its leading equivalent at $n = 2048$. These are finite-index diagnostics, not error bounds.

Regime	Parameters	Ratio
Critical fractional	$s = 2.5$, $u = U_c(s)$	0.997778596
Critical marginal	$s = 3$, $u = U_c(s)$	0.739119713
Critical finite curvature	$s = 4$, $u = U_c(s)$	1.002010230
Supercritical	$s = 2.5$, $b = 2A$	0.993099929
Supercritical integer	$s = 4$, $b = 2A$	1.016617966
Interior	$s = 2.5$, $u = 0.8U_c(s)$	1.001610430
Interior	$s = 3$, $u = 0.8U_c(s)$	1.000008400
Interior	$s = 4$, $u = 0.8U_c(s)$	0.999995456

The marginal row is deliberately retained. The leading order-three law converges slowly because the effective expansion parameter is $(\log\log n)/\log n$. At $n = 2048$ the coefficient is only 73.9% of its leading approximation. In contrast, the coefficient divided by the approximation through the second logarithmic correction in (4.14) is 1.010075657. The corresponding ratios at $n = 128$ and 512 are 1.049782812 and 1.020656682. This is useful evidence for the correction constants, while also showing why the leading equivalent alone can be misleading at computationally accessible indices.

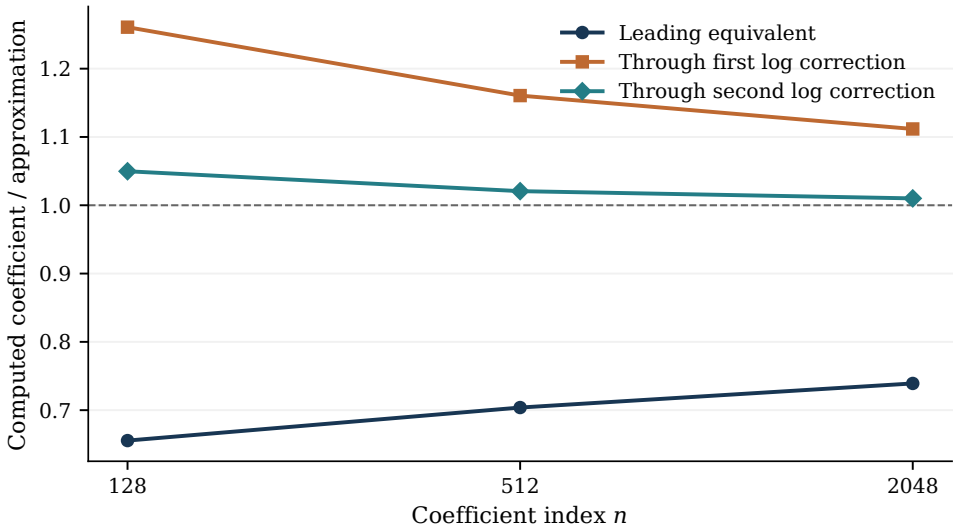


Figure 1: The order-three coefficient divided by successive asymptotic approximations. Points are exact finite recurrences in floating-point arithmetic; connecting segments only guide the eye. The horizontal line is ratio one. The three recorded indices are 128, 512, and 2048.

For the inverse charts, a separate 180-digit program solves the original polylogarithmic equation at $s = 2.5, 3, 3.5, 4$ and $d = 10^{-8}, 10^{-16}, 10^{-32}$. The largest residual, divided by d, is $1.89 \cdot 10^{-149}$ on this grid. The same program verifies the first core polynomials and the

transferred logarithmic corrections symbolically. A 200-digit program independently checks the critical order crossover for $\lambda = -2, 0, 2$ and $X = 20, 40, 80$. It evaluates both the original square-root deficit and the paired equation. At $X = 80$, the observed relative errors after the second algebraic correction range from $2.86 \cdot 10^{-6}$ to $7.12 \cdot 10^{-5}$. It also tests the exact first exponential coefficient in (5.18) against the difference between the full solution and the complete algebraic core. The separate joint-crossover program additionally tests nonzero η, checks the selected real branch against the interior fold, and includes a local complex-fold diagnostic; its complete parameters and residuals are in `data/joint_crossover_checks.json`. On the 18-point joint grid, the maximum absolute relative error of the first-corrected approximation decreases from $8.03 \cdot 10^{-3}$ at $X = 40$ to $2.15 \cdot 10^{-3}$ at $X = 80$. All six negative-η cases pass the independent fold-side check. At the complex test $k = 1, X = 40$, the full fold equations have tiny numerical residuals, but the leading limiting constants still differ from the computed α_c and ω_c by approximately 1.52 and 0.317, respectively. Solving the full equations accurately does not make their leading asymptotic approximation equally accurate.

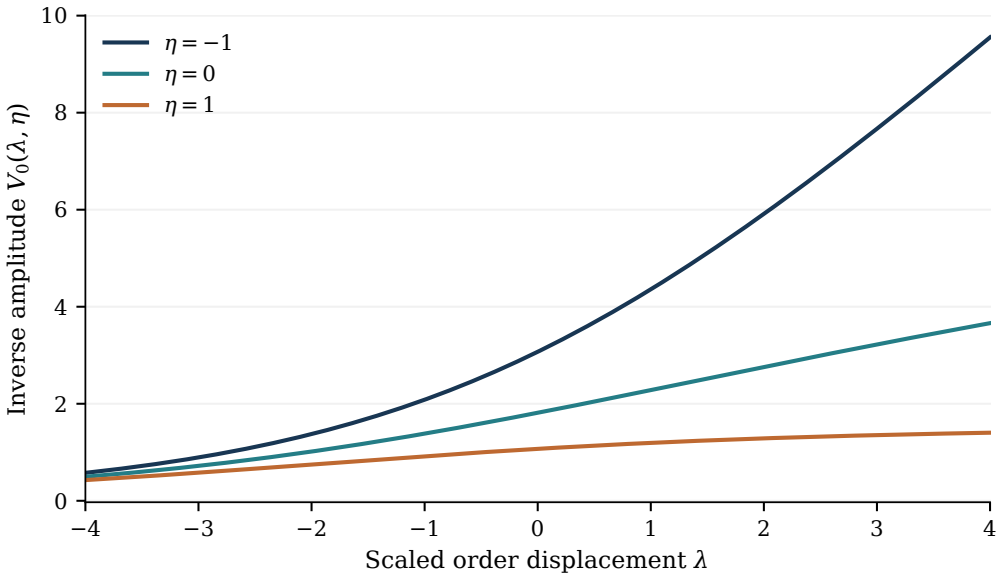


Figure 2: The explicit positive profile $V_0(\lambda, \eta)$ in (5.8). The curves are exact evaluations of the limiting formula, not fitted numerical data. They show how the order and feedback displacements jointly determine the inverse amplitude.

Finally, the coefficient-crossover program tests all 27 combinations of $s = 2.5, 3, 4$, $n = 128, 512, 2048$, and prescribed $\xi = -1, 0, 1$. The parameter b is selected on the locally unique branch near A, within $[A/2, 2A]$. This qualification matters: for $s > 3$ the map $b \mapsto \xi$ is not globally one-to-one on $(0, \infty)$. The stable density convention is independently checked by 75-digit Fourier integration at $\xi = \pm 1$ and by (6.15) at zero. The largest relative difference from the density evaluator is $5.15 \cdot 10^{-16}$.

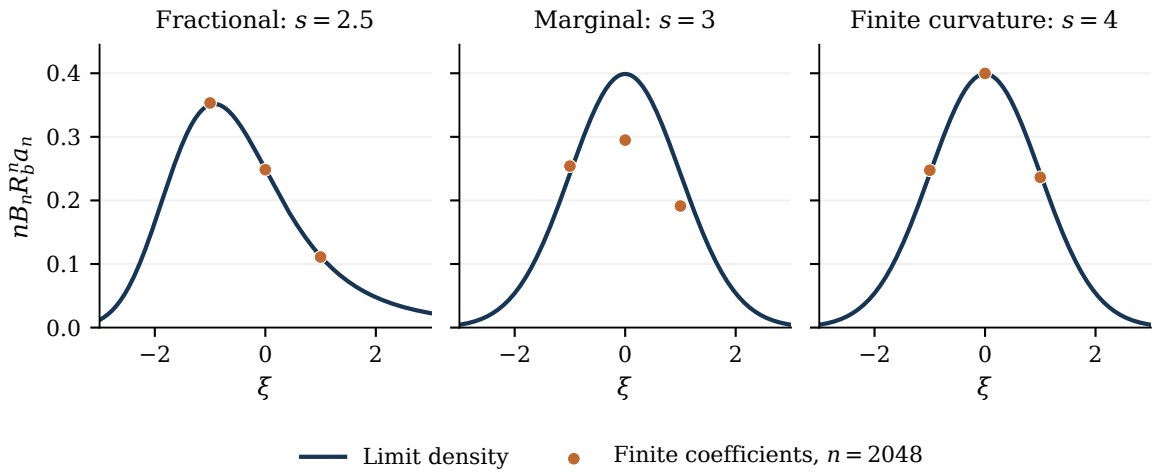


Figure 3: Coefficient crossover in the normalization $nB_nR_b^n a_n$. Curves are the stable or Gaussian limit densities; points are the finite coefficient computation at $n = 2048$ and $\xi = -1, 0, 1$. The elementary order-three normalization still has visible logarithmic error. Convergence need not be monotone at every noncentral ξ.

7.2 What should change in the inspected material

The source theorem for an interior characteristic point is correct in its stated range. Its boundary discussion was explicitly a proposed research problem, so the missing proof should not be described as an error. The natural source update is to mark its supercritical candidate as proved by Theorem 2.1, and replace its remaining critical-endpoint question by the three exact laws (2.5)–(2.7). The physical convergence statement can then be extended from $0 < s \leq 2$ to every $s > 0$ by Corollary 2.4.

The focused audit did not confirm a new false theorem in the inspected incoming reports. Several distinctions nevertheless need to survive integration, because omitting them would create incorrect extensions:

1. **The endpoint reference is not always the Taylor radius.** For $s > 2$ and $b > 0$, the equality $R = R_b$ holds at and above the feedback threshold. Below it, the physical germ first reaches an interior fold, whose radius is larger than the endpoint reference. The exact probability identity still uses R_b on both sides.

2. **Integer orders require the paired expression.** Gamma and zeta poles in the noninteger expansion cancel before any integer limit. For example the order-four term uses $H_3 = 11/6$, not a separate value of $\zeta(1)$. Similarly, the order-three constant $5/2$ includes the quadratic feedback contribution.

3. **A function chart and a coefficient expansion have different convergence claims.** The charts in Sections 3 and 5 are convergent holomorphic substitutions. The Gamma-derivative transfer series is used only to a fixed asymptotic order.

4. **Local complex folds need a sheet qualification.** Theorem 5.4 describes continued local normal-form solutions. It does not locate a singularity of the original Taylor germ without a separate global continuation argument.

5. **Uniformity requires stated parameter sets.** In the lattice limit, b stays in a compact positive interval. The compact ξ consequence then forces $b \to A$. The theorem does

not cover an unrelated sequence $b \to 0$, nor does an absolute local-limit error prove a relative tail estimate in an expanding ξ window.

The integration notes provide the source locations and a proposed cross-reference paragraph. They preserve the incoming source's historical theorem labels and credit the earlier linear-tail models. No existing theorem needs to be silently relabeled or removed.

7.3 Further research questions and concrete targets

The following questions are deliberately more specific than requests for "higher-order terms." Each identifies an additional statement that is not proved by the present theorems.

Q1. Joint coefficient asymptotics as both the order and feedback vary. Theorem 5.1 treats the local inverse with varying order; Theorem 6.2 keeps the order fixed. A natural next theorem would make the coefficient statement uniform when

$$s_n = 3 + \varepsilon_n, \qquad \lambda_n = \frac{\varepsilon_n \log n}{2}$$

stays bounded. Pole pairing suggests the candidate scale

$$\widetilde{B}_n^2 = \frac{n \log n}{2 b_n} \varphi(\lambda_n), \qquad \frac{n(1 - \zeta(s_n - 1)/b_n)}{\widetilde{B}_n} = O(1),$$

and a Gaussian limit in this normalization. This is a proposed target, not a theorem claimed here. The principal tasks are uniform Fourier domination through the colliding poles, the first correction, and comparison with the exact Lambert scale at $\lambda = 0$. The corresponding linear-tail results in [3] are a starting point, not a substitute for the nonlinear estimates.

Q2. Relative matching in expanding crossover windows. Determine the largest positive and negative ranges of ξ_n in which a relative version of (6.11) holds. On the positive side it should match the fixed supercritical n^{-s} law; on the negative side it should match the interior-fold exponential factor. Such a result requires a uniform large-deviation or saddle analysis beyond the absolute local limit.

Q3. Quantitative convergence radii for the charts. The holomorphic implicit function theorem supplies positive radii uniformly on compact parameter sets, but not optimal computable ones. Find certified polydiscs for the low-order, Lambert, and joint charts, and rigorous truncation constants that can be used numerically. The coefficient functions' possible pole at $w = 2$ is an algebraic clue; it is not already a determination of the chart's global boundary.

Q4. Global access to the complex fold curves. For fixed nonzero k, decide which paths in the complex order and feedback variables carry the original germ to the local branches of Theorem 5.4. Determine their monodromy and whether they ever become dominant singularities of a specified continuation. The local two-sheet normal form alone cannot answer this question.

Q5. All coefficient corrections away from the marginal order. Transfer the exact charts into finite formulas for every correction in the stable range and at higher orders, including cancellations when different chart monomials produce the same power. At rational orders, distinguish merging exponents from zeros of reciprocal Gamma factors. The conditional disappearance of the ladder at $\zeta(s - 2) = -1$ gives a particularly clean test case.

Q6. General analytic feedback potentials. Replace $uv - v^2/2$ by an analytic $F(v)$ with $F(0) = 0$ and study $F(v) = \mathrm{Li}_s(re^v)$. For $s > 3$, if $F(v_b) = \zeta(s)$ and $F'(v_b) = \zeta(s-1)$, the finite-curvature term is governed by $\zeta(s-2) - F''(v_b)$. Find hypotheses preserving positive coefficients and global dominance, and classify what happens when this curvature is tuned to zero. The latter can create a higher degeneracy and needs a different chart.

Q7. Logarithmically weighted and slowly varying tails. Replace k^{-s} by $k^{-s}(\log k)^q$, or by $k^{-s}L(k)$ under explicit regularity assumptions. Order derivatives of the polylogarithm give exact finite logarithmic weights, linking this problem to the incoming order-jet articles. Determine which tails still admit convergent finite-generator charts and which yield only asymptotic scales. A regular-variation equivalent alone cannot imply an analytic convergent chart.

Q8. Arithmetic support and several dominant points. The uniqueness proof uses the positivity of every coefficient. For feedback supported on a proper lattice, classify the competing root-of-unity singularities and the resulting periodic coefficient laws. More generally, identify the least support hypotheses under which the strict triangle argument and aperiodic local limit remain valid.

Q9. Finite action cutoffs in the nonlinear kernel. Study truncations before and after the nonlinear transformation $f_u(t) = u - \sqrt{u^2 - 2\,\mathrm{Li}_s(t)}$. These are different operations. Determine the cutoff required to preserve each critical coefficient law and the crossover profile, and whether the prefix cancellations in the linear endpoint literature survive the quadratic feedback.

Q10. Certified and formal verification. Convert the finite coefficient recursions and Lagrange extraction identities into proof-assistant statements first. An interval implementation of the sector bounds and Rouché circles could then certify explicit numerical domains. Formalizing the global continuation and logarithmic transfer steps is a separate, larger project; the supplied numerical scripts do not accomplish it.

Questions Q1–Q3 offer the most direct continuation of the present analysis. Q4 and Q6 address genuinely additional analytic structure; Q7–Q9 connect the exact model to broader tail classes and the action budgets already studied in the repository. The existing finite identities, positivity argument, and pole-paired normal form provide concrete starting points for each direction.

7.4 Reproducibility and integration

The archive contains the master `article.tex`, its seven section files, the bibliography source, the compiled PDF, five verification programs, a figure-generation program, machine-readable records, and integration notes. The recorded data permit the figures and PDF to be rebuilt without rerunning high-precision roots. The optional verification target recomputes all numerical and symbolic checks. The README gives the commands, dependency versions, and a distinction between exact arithmetic checks and numerical diagnostics. The source provenance pins the repository revision and hashes the specific comparison files and incoming archives.

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[2] *Beyond Finite-Action Folds: Critical Hahn Transseries, Stable Sector Asymptotics, and Sharp Action Budgets*. Research draft prepared for Vladimir Reshetnikov, 29 September 2026. ProveIt, `Analysis/Transseries/docs/series-and-transseries/Critical_Hahn_Transseries_Beyond_Finite_Action_Folds/article.tex`. In particular Proposition `prop:exact`, Theorem `thm:phases`, and Proposition `prop:local-crossover`.

[3] *Through the Stable–Gaussian Endpoint: Uniform Critical Transseries, Prefix-Independent Cutoff Corrections, and Conditional Extremes*. Research article prepared for Vladimir Reshetnikov, 29 September 2026. ProveIt, `Analysis/Transseries/docs/series-and-transseries/Stable_Gaussian_Endpoint_Uniform_Critical_Transseries/article.tex`.

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